

Komatsu Parts Procurement & SAP Business One PO Automation

Operational Fleet Context, Komatsu PDX Split Rules, and Unified SAP B1 Service Layer Integration Specification

TARGET AUDIENCE	PREPARED BY	VENDOR MASTER	DOCUMENT VERSION
ABS (SAP B1 Technical Team)	Motasem Ghanem (EQP Ops)	Komatsu Middle East (V000006)	v2.2 (Enterprise RFC Specification)

1. Operational Fleet Context: Mission-Critical Asset Management

Dar Al Hai General Trading operates a premier fleet of heavy earthmoving and construction equipment deployed across major civil infrastructure, deep excavation, and coastal development projects throughout Kuwait. The operational core includes heavy **Komatsu hydraulic excavators** (such as PC500LC-10R and PC400), crawler bulldozers (D375A), and high-capacity wheel loaders (WA470). In continuous 24/7 operating environments, equipment availability directly governs project schedules, and unbudgeted downtime creates severe financial penalties and contractual delivery risks.

To sustain maximum machine availability, the Maintenance & Fleet Logistics division procures genuine OEM replacement components directly from the authorized regional distributor entity, **Komatsu Middle East FZE (KME)**, utilizing their digital B2B distributor portal (**Komatsu PDX**). In SAP Business One, Komatsu Middle East is configured as foreign Vendor **V000006**, transacting strictly in **US Dollars (USD)**.

2. Core Architectural Principle: Absolute Uniformity for SAP Business One

Guiding Engineering Tenet for ABS Consultants & Developers:

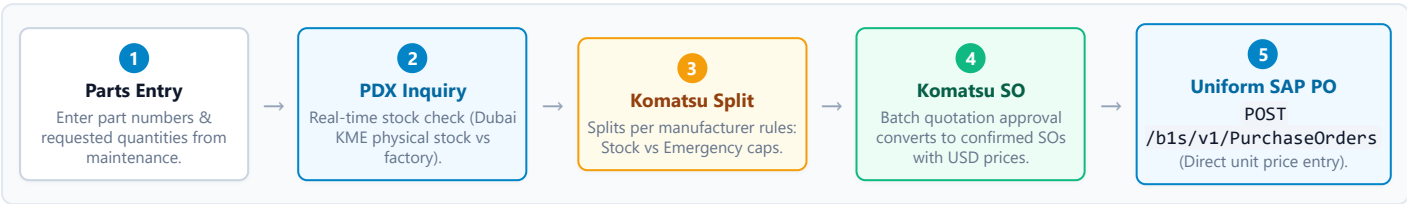
To SAP Business One, all incoming transactions are uniform, standard Purchase Orders (OPOR).

Regardless of whether parts were allocated to routine warehouse replenishment or urgent machine breakdown repairs, and regardless of how orders were partitioned upstream on the manufacturer's portal, **the SAP integration schema, business object, and financial posting process remain 100% identical:**

- Native Manufacturer Pricing (Automatic Passthrough):** The unit price generated by Komatsu PDX on the confirmed Sales Order represents the final, binding price. For Emergency Orders (**EO**), Komatsu automatically embeds emergency rates directly into the line-item unit price. For Daily Stock (**DS**), it applies standard catalogue distributor rates.
- Zero Surcharge Calculation in Middleware or SAP:** Neither SAP Business One nor the API middleware calculates surcharges, markups, freight premiums, or custom pricing adjustments. SAP simply records the exact confirmed unit price received from Komatsu.
- Deterministic 1:1 Document Mapping:** Every confirmed Komatsu Sales Order (SO) creates exactly one standard Purchase Order in SAP B1 via the Service Layer REST API (POST /b1s/v1/PurchaseOrders).

3. End-to-End System Architecture Flow

The workflow establishes an automated, auditable chain of custody from field technician requisition to SAP general ledger commitment:



Strict Separation of Concerns

The EQP system handles all Komatsu OEM logistics logic, distributor sequencing (R.../2026), and multi-chassis quota allocation upstream, shielding SAP B1 from operational complexity.

Auditability & Ledger Integrity

By capturing native manufacturer pricing and vendor order identifiers at origin, every purchase order maintains unbroken traceability through warehouse receipt to financial invoice settlement.

Summary for ABS Integration Engineers:

SAP Business One acts exclusively as the authoritative financial ledger and inventory repository. No custom business logic, document types, or price recalculation rules are required inside SAP B1.

4. Operational Constraint: Why Orders Are Split on Komatsu PDX

In standard enterprise ERP configurations, ordering 50 units of a spare part is recorded as a single purchase requisition line. However, the **Komatsu Middle East PDX Portal** enforces strict manufacturer routing and anti-hoarding policies designed to safeguard factory inventory across the Middle East. Consequently, our EQP procurement middleware must partition requisitions into distinct sub-orders before submission.

Crucial Distinction for the ABS Consulting Team:
Order splitting is strictly an external Komatsu manufacturer policy constraint, NOT an internal SAP data modeling requirement. Sub-order generation and machine allocation exist solely to comply with Komatsu PDX portal validation rules prior to order submission. Once Komatsu approves and confirms the sub-orders, each Sales Order maps cleanly 1:1 into standard SAP Purchase Orders.

Fulfillment Channels Comparison: Normal Daily Stock (DS/SO) vs. Emergency Breakdown (EO)

OPERATIONAL DIMENSION	NORMAL DAILY STOCK (DS / SO)	EMERGENCY BREAKDOWN (EO)
Physical Inventory Sourcing	Fulfilled directly from KME Dubai central distribution warehouse inventory.	Sourced from Komatsu Ltd. (KLTD) Japan factory or global emergency reserve.
Ordering Quotas & Caps	Uncapped bulk ordering. No maximum line-item limits imposed by Komatsu.	Strictly capped per chassis serial. Komatsu restricts maximum quantity per machine (e.g., max 20 units/chassis) to prevent distributor stockpiling.
Equipment & Asset Binding	No machine or customer required. Treated as general inventory replenishment.	Mandatory equipment binding. Must specify verified machine model (e.g. PC500LC-10R) and valid chassis serial (e.g. 100433).
Unit Pricing Mechanism	Base catalogue distributor rate, computed natively by Komatsu PDX.	Emergency rate, computed natively by Komatsu PDX directly on the unit price.
Distributor Reference	Sequential distributor reference number (e.g., R229/2026).	Sequential distributor reference numbers (e.g., R230/2026 , R231/2026).

5. Concrete Walkthrough: Requisition for 50 EA Fuel Filters (Part 600-311-6742)

When field maintenance requests 50 fuel filters, the EQP system checks live PDX inventory and automatically executes a hybrid split across channels:

Requisition: 50 EA • Fuel Filter (600-311-6742)

Komatsu PDX Status: KME Dubai Stock: 10 EA | Factory Shortage: 40 EA

- Sub-Order #1 (DS Stock) • Ref R229/2026 : 10 units fulfilled from Dubai physical stock → No machine required → Converted to Komatsu SO #0000278046 at confirmed catalogue price → Triggers SAP PO #1.
- Sub-Order #2 (EO Emergency) • Ref R230/2026 : 20 units assigned to Excavator Chassis 100433 (capped at 20) → Converted to Komatsu SO #0000278047 at confirmed PDX price → Triggers SAP PO #2.
- Sub-Order #3 (EO Emergency) • Ref R231/2026 : 20 units assigned to Excavator Chassis 100434 (capped at 20) → Converted to Komatsu SO #0000278048 at confirmed PDX price → Triggers SAP PO #3.

6. Financial Reconciliation & Three-Way Matching Governance

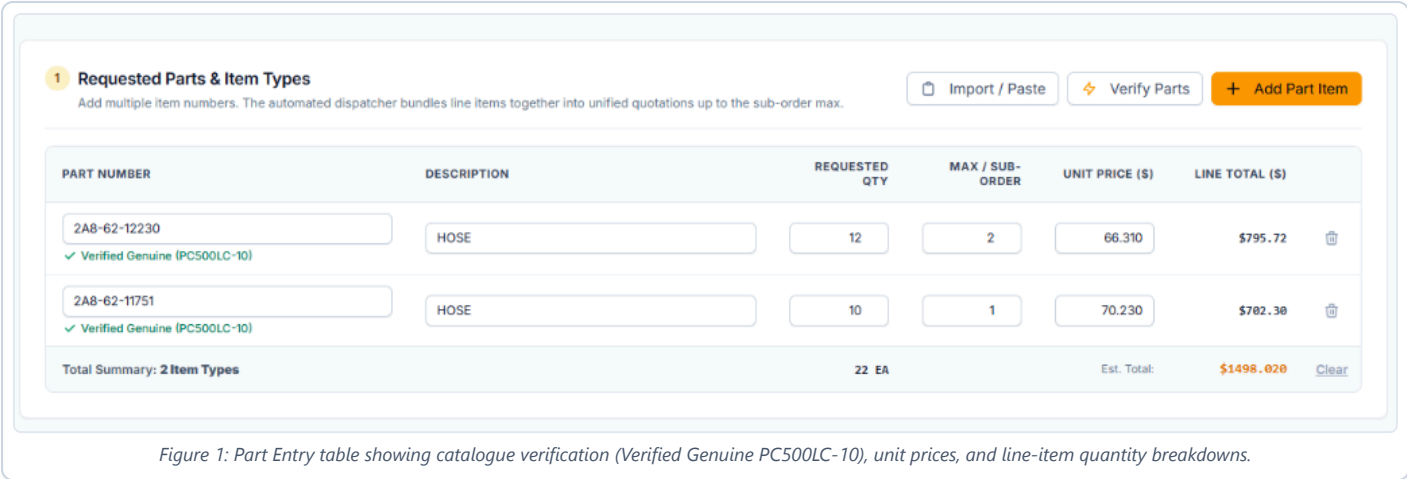
Flawless Accounting & Warehouse Reconciliation in SAP Business One:
Each resulting SAP Purchase Order stores the exact Komatsu Sales Order Number in NumAtCard and U_KomatsuSO .
When Komatsu Middle East ships the goods, their shipping documents and commercial invoices correspond 1:1 with individual Komatsu SOs:

- **Goods Receipt PO (GRPO):** Warehouse staff receive incoming parts in SAP B1 against the specific PO referencing Komatsu's shipping packing slip and delivery note.
- **A/P Invoice 3-Way Matching:** Accounts Payable matches Komatsu's USD commercial invoice against the GRPO and PO with zero price variance, because SAP recorded the exact confirmed unit price certified by Komatsu.
- **Currency Realization:** Foreign currency liability is recognized at transaction time, eliminating downstream ledger adjustments and manual credit memo reconciliations.

7. Operational Workflow Phase I: Part Entry & Fleet Allocation

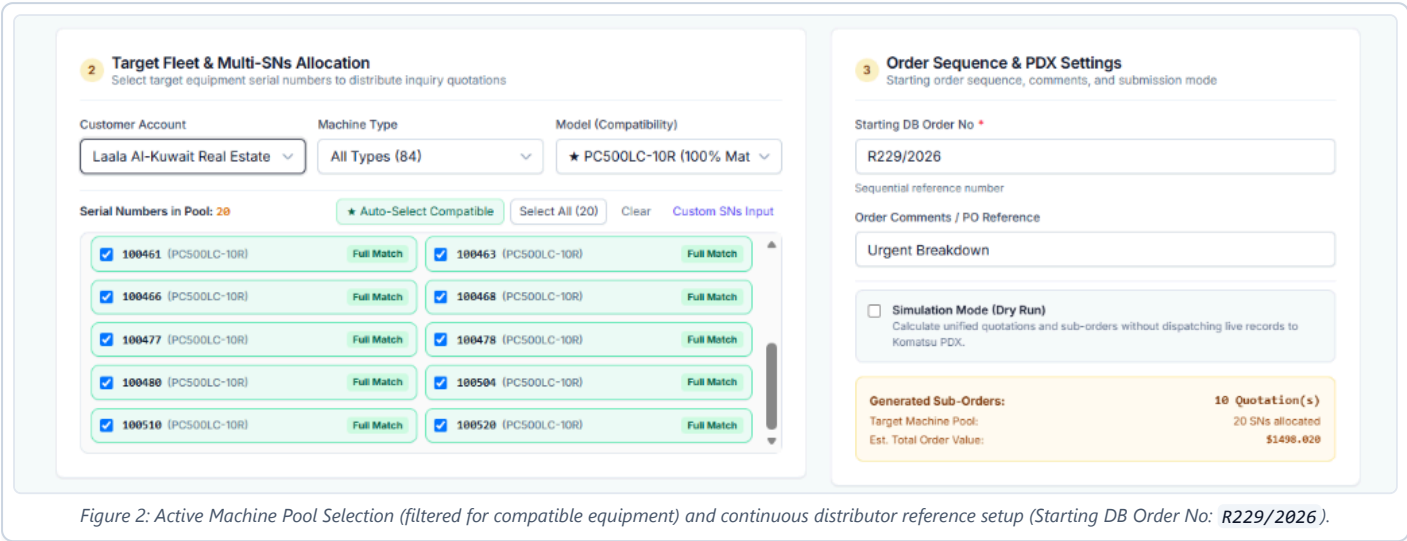
Step 1: Part Entry, Real-Time Master Stock & Allocation

The maintenance engineer enters requested part numbers. The EQP engine verifies them against the Komatsu catalogue master, displays live inventory badges (**KME Stock** in Dubai, **EOR** emergency restriction, **KLTD** factory stock), and calculates available quantities versus emergency shortage.



Step 2: Target Fleet & Multi-SN Allocation (Emergency Orders Only)

When items route through the Emergency breakdown channel, the dispatcher selects the target customer account (e.g. *Laala Al-Kuwait Real Estate Co.*) and selects compatible heavy equipment chassis serials (e.g. *PC500LC-10R* pool). For 100% Stock orders, machine allocation is completely bypassed.



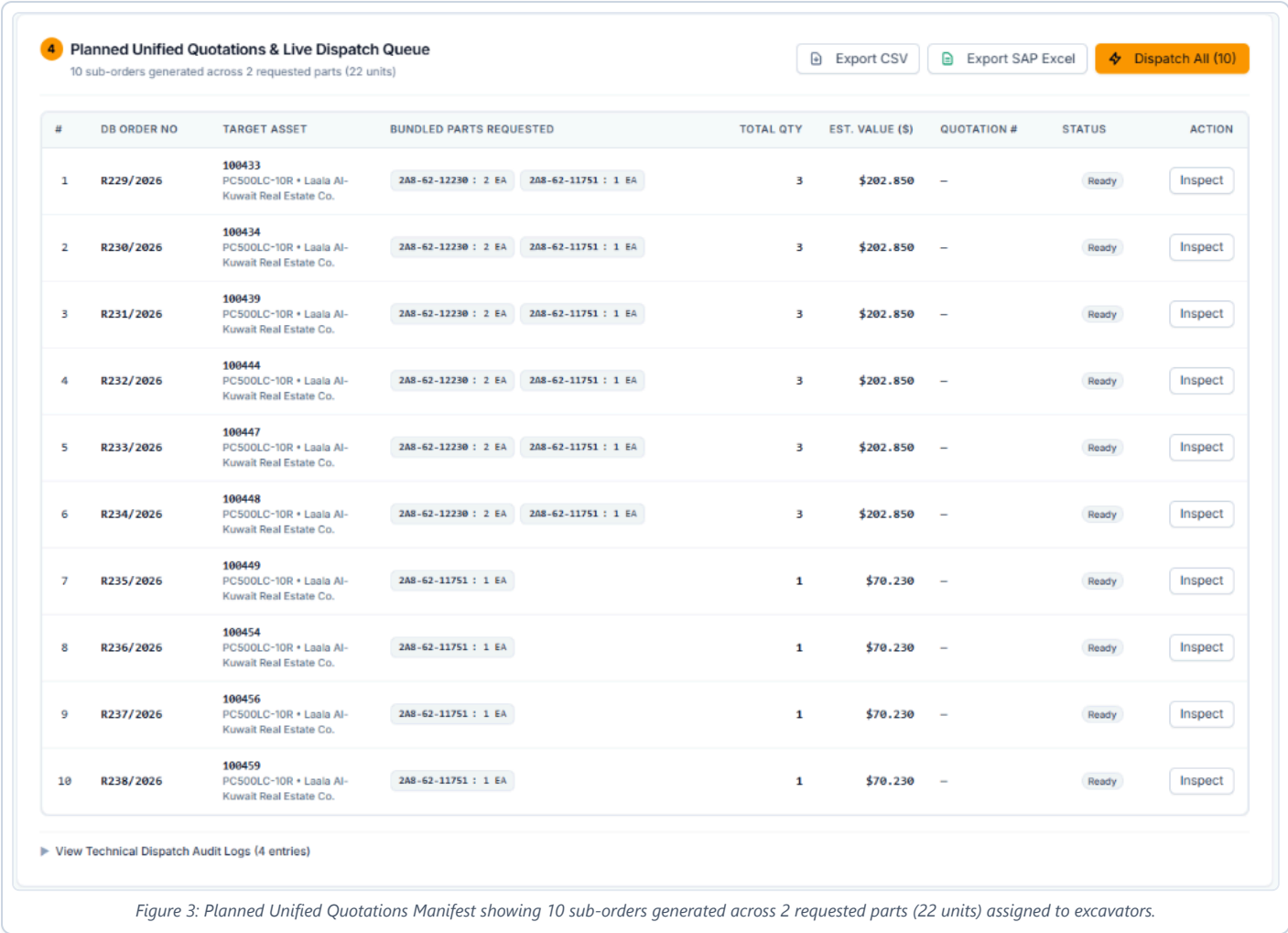
Automated Dispatch Preparation & Dry-Run Simulation:

The system calculates estimated quotation counts and order values in real-time, allowing operators to run dry-run simulations before dispatching live records to Komatsu PDX. This guarantees full data validation prior to external transmission.

8. Operational Workflow Phase II: Planned Unified Quotations Manifest

Step 3: Planned Unified Quotations Manifest & Live Dispatch Queue

The EQP splitting engine bundles line items across assigned excavator chassis into unified distributor quotations within Komatsu sub-order caps. Each sub-order receives a sequential distributor reference number (R229/2026 through R238/2026) ensuring continuous audit numbering.



Queue Structure & Dispatch Governance Controls

QUEUE ATTRIBUTE	OPERATIONAL IMPLEMENTATION	GOVERNANCE & AUDIT FUNCTIONALITY
DB Order Number	Sequential format: R<Seq>/2026	Ensures non-gapping, continuous audit tracking across distributor submissions.
Target Asset Binding	Chassis SN + Model + Customer Account	Enforces Komatsu's manufacturer requirement for emergency breakdown allocation.
Bundled Part Lines	Multi-part bundling per quote (e.g. 2 Hoses)	Maximizes shipping efficiency and minimizes individual sub-order transaction count.
Export & Audit Tools	CSV Export • SAP Excel Export • Audit Logs	Provides immediate offline backup and full dispatch traceability for procurement supervisors.

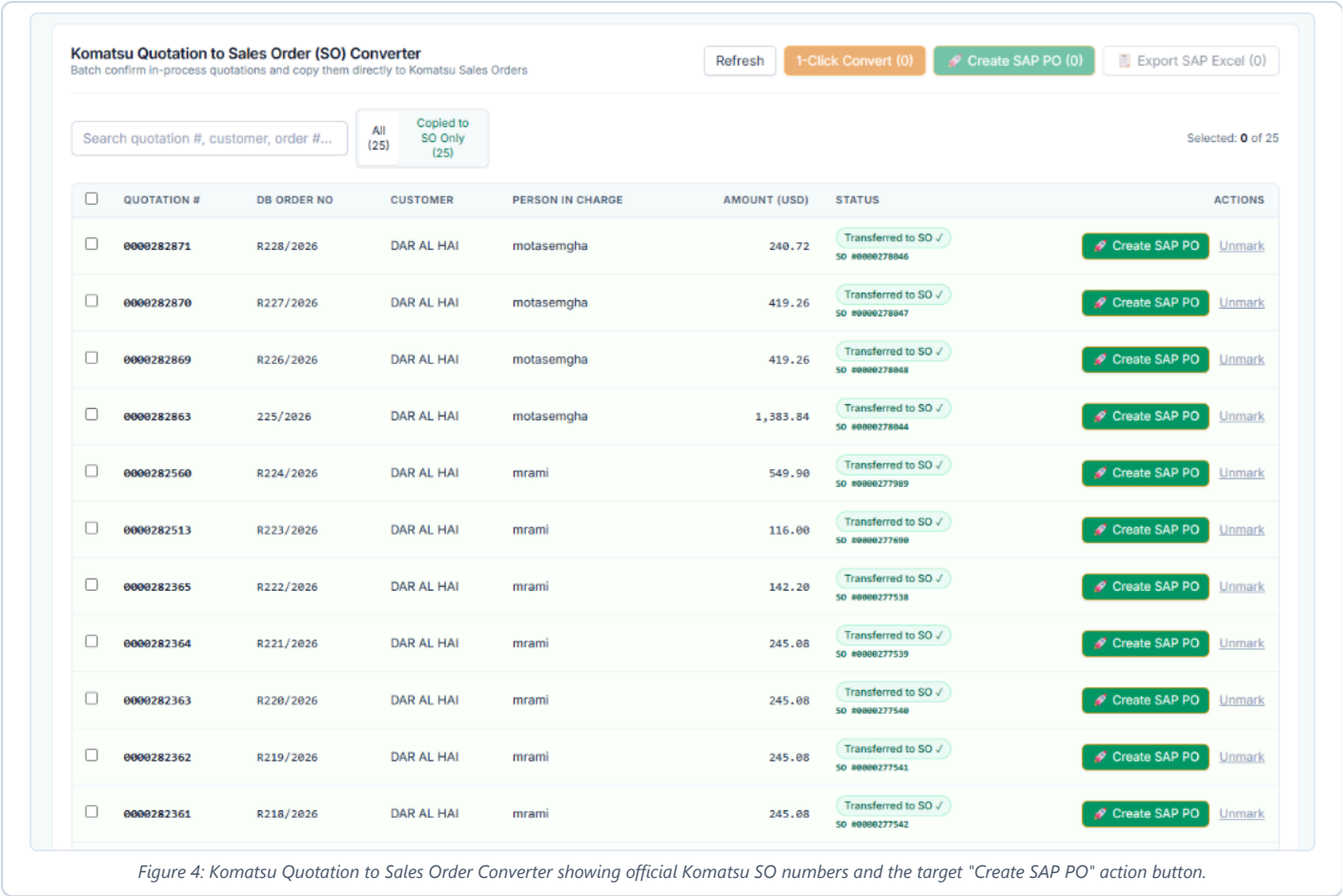
Batch Dispatch Resilience:

The live dispatch queue provides granular inspection of each sub-order prior to execution, preventing accidental submissions and ensuring strict compliance with Komatsu Middle East ordering quotas.

9. Operational Workflow Phase III: Quotation-to-SO Conversion & SAP Trigger

Step 4: Komatsu Quotation to Sales Order (SO) Converter

Once submitted to Komatsu PDX, the portal confirms the sub-orders with official Komatsu Quotation Numbers (e.g. 0000282871). The EQP procurement interface enables 1-Click batch conversion to certified **Komatsu Sales Orders (SO)** (e.g. SO #0000278046) carrying binding USD purchase prices.



The Exact SAP Integration Execution Trigger

Direct Service Layer Execution Lifecycle:

- At this stage, the Komatsu Sales Order is officially confirmed with certified USD pricing. Clicking "Create SAP PO" initiates the following automated sequence:
- Payload Serialization:** Middleware packages confirmed part numbers, quantities, certified unit prices, and tracking references into a standard Service Layer JSON document.
 - Service Layer Call:** Sends POST /b1s/v1/PurchaseOrders to SAP Business One.
 - Response & Persistence:** On HTTP 201 Created, SAP returns the official DocNum (e.g., 20260481) and DocEntry, which are permanently written to the database.
 - Idempotency Lock:** The button updates to "SAP PO #20260481 ✓" and locks permanently, eliminating duplicate document creation.

Error Handling & Retry Resilience:

In the event of network timeouts or temporary Service Layer unavailability, the system captures full diagnostic logs, preserves the quotation state, and permits single-click retry without risk of duplicate PO creation.

10. Technical API Specification for the ABS / SAP Team

Direct creation of Purchase Orders in SAP Business One will be executed via the **SAP Business One Service Layer** (REST / OData v4) over HTTPS (TLS 1.3), utilizing standard session-based authentication (`B1SESSION` + `ROUTEID`).

A. Header Field Mapping (Table: `OPOR` / Entity: `PurchaseOrders`)

SAP FIELD	DB FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
CardCode	CardCode	Vendor Code	Komatsu Middle East Vendor Code	"V000006"
DocDate	DocDate	Posting Date	Order Date (YYYY-MM-DD)	"2026-10-05"
DocDueDate	DocDueDate	Delivery Due Date	Required Delivery Date	"2026-10-12"
DocCurrency	DocCur	Document Currency	Komatsu Billed Currency	"USD"
NumAtCard	NumAtCard	BP Reference No	Komatsu SO Number + DB Ref	"SO #0000278046 / R229/2026"
Comments	Comments	Document Remarks	Order context & reference notes	"Komatsu Ref: R229/2026 Quote: 0000282871"
U_KomatsuSO	U_KomatsuSO	UDF: Komatsu SO No	Official Komatsu Sales Order No	"0000278046"
U_KomatsuQuote	U_KomatsuQuote	UDF: Komatsu Quote No	Official Komatsu Quotation No	"0000282871"
U_MachineSerial	U_MachineSerial	UDF: Machine Serial	Asset Serial (if EO) or blank (Stock)	"100433" (or null)
U_MachineModel	U_MachineModel	UDF: Machine Model	Equipment Model (if EO) or blank	"PC500LC-10R" (or null)

B. Line Item Field Mapping (Table: `POR1` / Entity: `DocumentLines`)

SAP FIELD	DB FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
ItemCode	ItemCode	Item Number	Genuine Komatsu Part Number	"600-311-6742"
ItemDescription	Dscription	Item Description	Part Description from Komatsu Master	"FUEL FILTER"
Quantity	Quantity	Quantity	Confirmed order line quantity	10.00
UnitPrice	Price	Unit Price (USD)	Confirmed unit price directly from Komatsu SO	25.00
WarehouseCode	WhsCode	Warehouse	Receiving Warehouse	"003" (or per ABS)
TaxCode	TaxCode	Tax Code	Import Zero / Exempt Tax Code	"P0" (or per ABS)

Currency Valuation & Foreign Exchange Integrity:

Specifying `DocCurrency`: "USD" ensures SAP Business One records vendor liability in foreign currency (USD) while automatically calculating book values in Kuwaiti Dinar (KWD) based on SAP's official daily exchange rate table (`ORTT`). Financial ledger integrity is preserved without custom scripting.

Session Management & Authentication Protocol:

The middleware performs automated session login via `POST /b1s/v1/Login` , manages HTTP cookies (`B1SESSION` and `ROUTEID`), pools active sessions across requests, and automatically handles re-authentication if an expired session token is encountered.

11. Service Layer Payload Contract & Item Master Protocol

Standard Request (POST /b1s/v1/PurchaseOrders) & Response Contract

```
{
  "CardCode": "V000006",
  "DocDate": "2026-10-05",
  "DocDueDate": "2026-10-12",
  "DocCurrency": "USD",
  "NumAtCard": "SO #000278046 / R229/2026",
  "Comments": "Komatsu Order | DB Ref: R229/2026 | Quote: 000282871",
  "U_KomatsuSO": "000278046",
  "U_KomatsuQuote": "000282871",
  "U_MachineSerial": "100433",
  "U_MachineModel": "PC500LC-10R",
  "DocumentLines": [
    {
      "ItemCode": "600-311-6742",
      "ItemDescription": "FUEL FILTER",
      "Quantity": 10.0,
      "UnitPrice": 25.00,
      "WarehouseCode": "003",
      "TaxCode": "P0"
    }
  ]
}
```

Expected SAP Response (HTTP 201 Created):

```
{
  "DocEntry": 14205,
  "DocNum": 20260481,
  "DocDate": "2026-10-05",
  "CardCode": "V000006",
  "DocTotal": 250.00,
  "DocCurrency": "USD"
}
```

Item Master Data Handling (OITM):

With over 500,000 Komatsu part numbers, occasionally a requested item is not yet registered in SAP B1. The EQP middleware checks `GET /b1s/v1/Items('{ItemCode}')` prior to PO creation. If HTTP 404 is returned, our API automatically provisions the item via `POST /b1s/v1/Items` using verified Komatsu catalogue data before dispatching the PO.

12. Actionable Alignment Checklist & RFC Questions for ABS

1. Service Layer Endpoint URL & Connectivity:

Please provide the official Service Layer URL (e.g. <https://sap.daralhai.com:50000/b1s/v1/>) and confirm firewall whitelisting for the EQP backend IP.

2. Dedicated Technical API User:

We request a dedicated API service user (e.g. `B1_API_EQP`) with permissions restricted to `PurchaseOrders` (Create/Read) and `Items` (Read/Create) to avoid reliance on named user accounts.

3. Handling New Part Numbers (Item Master OITM):

With over 500,000 Komatsu parts in the catalogue, occasionally a component does not yet exist in SAP B1. What is ABS's preferred procedure?

- *Option 1 (Recommended):* Allow our API to auto-create missing parts via `POST /b1s/v1/Items` using Komatsu catalogue master data before creating the PO.
- *Option 2:* Use a standard non-inventory item master with explicit line-item descriptions.

4. Document Status Workflow (Active PO vs. Draft):

Should the API create approved open POs (`PurchaseOrders`) directly, or create Drafts (`PurchaseOrderDrafts`) for accounting verification during the initial sandbox pilot?

5. Warehouse & Tax Defaults Confirmation:

Please confirm that receiving Warehouse Code `003` and Tax Code `P0` (0% import tax) are the appropriate system defaults for Vendor `V000006`.

6. User-Defined Field (UDF) Setup:

Please confirm if user-defined fields (`U_KomatsuSO`, `U_KomatsuQuote`, `U_MachineSerial`, `U_MachineModel`) are already configured on `OPOR`, or if ABS will provision them.

Testing SLA & Next Steps:

Upon receipt of ABS sandbox credentials, EQP engineering will execute end-to-end test transactions within 24 hours to validate bidirectional document generation and trace logging.